VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Configure IP ACLs to Mitigate Attacks

Prepared by:
Kudzaishe Majeza
Junior Network Engineer – VxLEARN Networks

Mentor:
Titus Majeza
Senior Network Engineer

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Introduction

In this activity, access control lists (ACLs) are configured to meet network communication security goals. Access to routers R1, R2, and R3 is restricted to the management station PC-C, and additional ACLs are applied to edge routers R1 and R3 to mitigate common network attacks based on source and destination IP address.

Objectives

- Verify connectivity among devices before firewall (ACL) configuration.
- Use ACLs to ensure remote access to the routers is available only from management station PC-C.
- Configure ACLs on R1 and R3 to mitigate attacks.
- Verify ACL functionality.

Background / Scenario

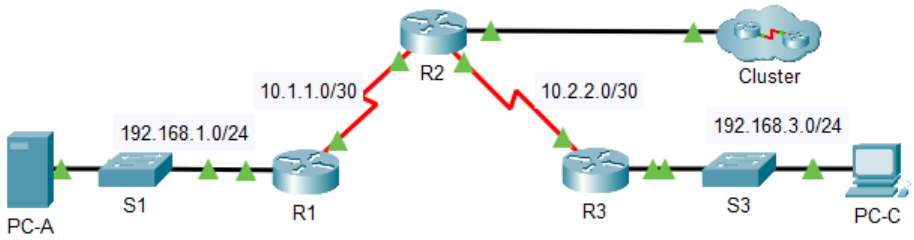
Access to routers R1, R2, and R3 should only be permitted from PC-C, the management station. PC-C is also used for connectivity testing to PC-A, which is a server providing DNS, SMTP, FTP, and HTTPS services.

Standard operating procedure is to apply ACLs on edge routers to mitigate common threats based on source and destination IP address. In this activity, ACLs are created on edge routers R1 and R3 to achieve this goal. ACL functionality is then verified from internal and external hosts.

The routers have been pre-configured with the following:

- Enable password: ciscoenpa55
- Password for console: ciscoconpa55
- SSH logon username and password: SSHadmin / ciscosshpa55
- IP addressing: pre-configured
- Static routing: pre-configured

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/1	192.168.1.1	255.255.255.0	N/A
R1	S0/0/0	10.1.1.1	255.255.255.252	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A
R2	S0/0/1	10.2.2.2	255.255.255.252	N/A
R2	G0/0	209.165.200.225	255.255.255.224	N/A
R2	Lo0	192.168.2.1	255.255.255.0	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A
R3	S0/0/1	10.2.2.1	255.255.255.252	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1

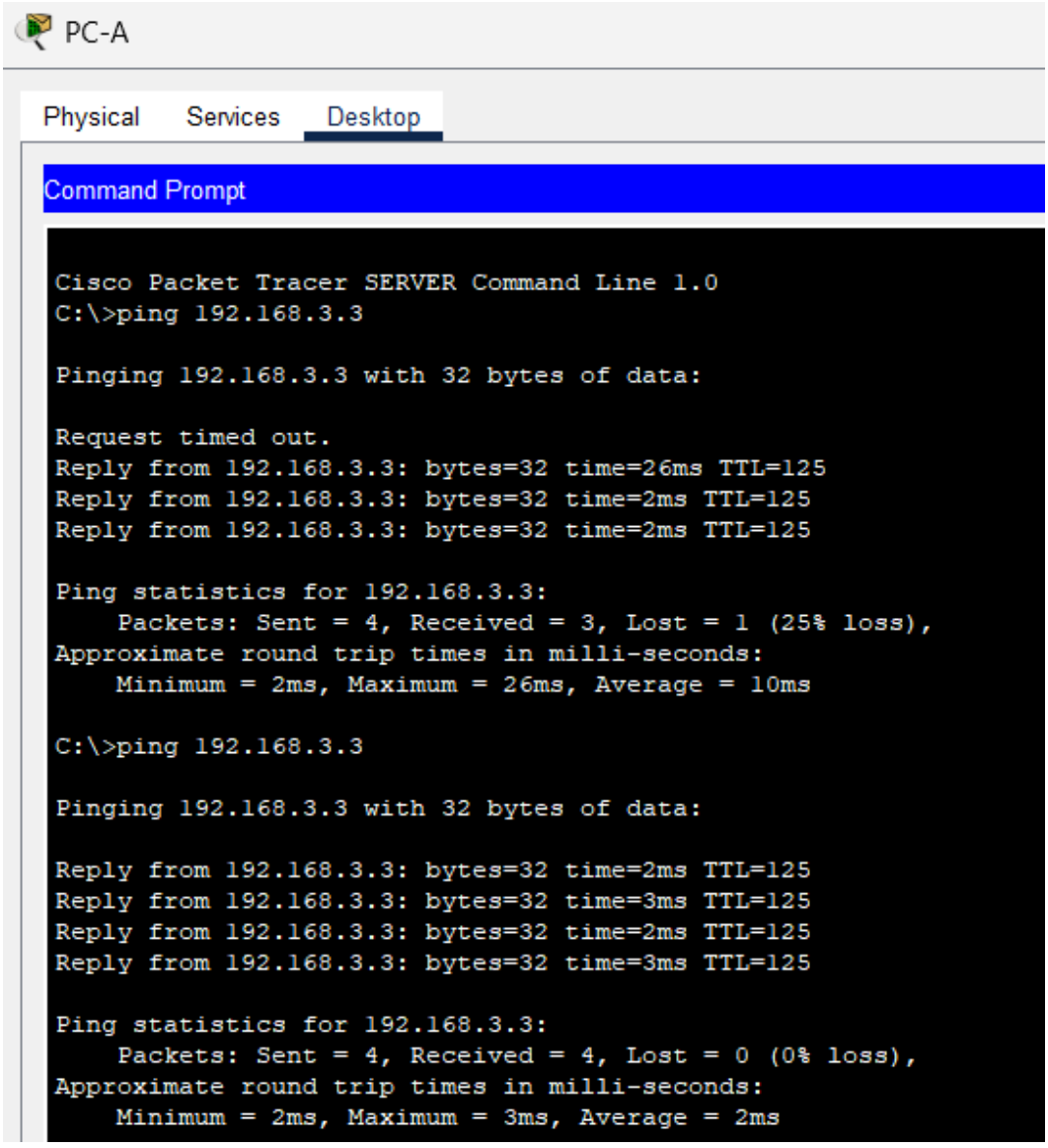
Part 1: Verify Basic Network Connectivity

Verify network connectivity prior to configuring the IP ACLs.

Step 1: From PC-A, verify connectivity to PC-C and R2.

a. From the PC-A command prompt, ping PC-C (192.168.3.3).

```
C:\> ping 192.168.3.3
```



Observation / Result: First ping to 192.168.3.3 showed 1 timeout (25% loss) due to initial ARP resolution; second ping succeeded with 0% loss (4/4 replies, avg 2ms), confirming end-to-end connectivity between PC-A and PC-C.

b. From the PC-A command prompt, establish an SSH session to the R2 Lo0 interface (192.168.2.1) using username SSHadmin and password ciscosshpa55. Close the session when finished.

```
PC-A> ssh -l SSHadmin 192.168.2.1
```

```
C:\>ssh -l SSHadmin 192.168.2.1

Password:
% Login invalid


Password:



R2#
```

Observation / Result: The first SSH attempt returned "% Login invalid" due to a password entry error; on the second attempt with the correct password (ciscosshpa55), the session succeeded and reached the R2 privileged prompt (R2#), confirming SSH access to R2's Lo0 interface works correctly.

Step 2: From PC-C, verify connectivity to PC-A and R2.

a. From the PC-C command prompt, ping PC-A (192.168.1.3).

```
C:\> ping 192.168.1.3
```

PC-C

Physical Desktop

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=36ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125

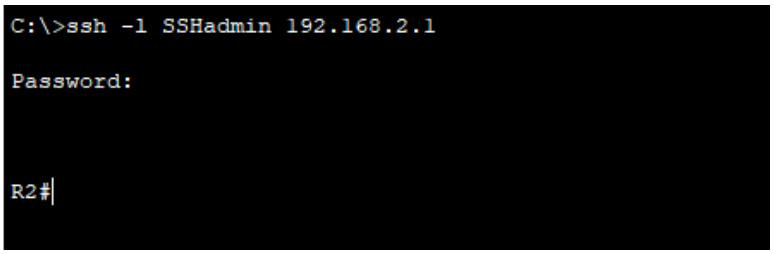
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 36ms, Average = 10ms

C:\>
```

Observation / Result: Ping to 192.168.1.3 succeeded with 4/4 replies and 0% loss (avg 10ms, max 36ms), confirming end-to-end connectivity between PC-C and PC-A.

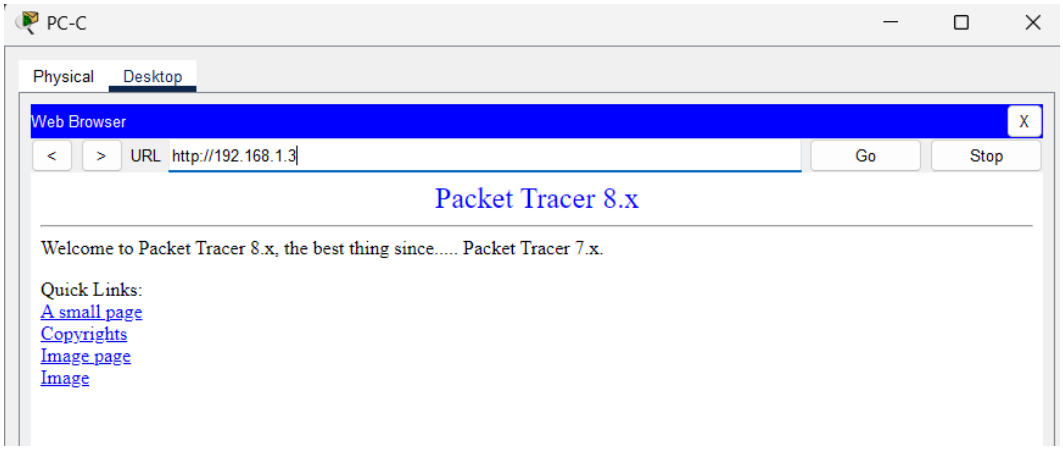
b. From the PC-C command prompt, establish an SSH session to the R2 Lo0 interface (192.168.2.1) using username SSHadmin and password ciscosshpa55. Close the session when finished.

```
PC-C> ssh -l SSHadmin 192.168.2.1
```



Observation / Result: The SSH session from PC-C to 192.168.2.1 succeeded on the first attempt using username SSHadmin and password ciscosshpa55, reaching the R2 privileged prompt (R2#), confirming SSH access to R2's Lo0 interface works correctly.

c. Open a web browser on PC-C to the PC-A server (192.168.1.3) to display the web page. Close the browser when done.



Observation / Result: PC-C successfully loaded the default web page hosted on PC-A (http://192.168.1.3), confirming HTTP connectivity end-to-end before any ACLs were applied. An initial attempt returned a connection reset, but the page loaded successfully on retry.

Part 2: Secure Access to Routers

Step 1: Configure ACL 10 to block all remote access to the routers except from PC-C.

Use the access-list command to create a numbered IP ACL on R1, R2, and R3.

```
R1(config)# access-list 10 permit host 192.168.3.3
R2(config)# access-list 10 permit host 192.168.3.3
R3(config)# access-list 10 permit host 192.168.3.3
```

```
R1>enable
Password:
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 10 permit host 192.168.3.3
R1(config)#show access-lists 10
           ^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show access-lists 10
Standard IP access list 10
    permit host 192.168.3.3

R1#
```

```
R2>enable
Password:
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#access-list 10 permit host 192.168.3.3
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show access-lists 10
Standard IP access list 10
    permit host 192.168.3.3

R2#
```

```
R3>enable
Password:
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#access-list 10 permit host 192.168.3.3
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show access-lists 10
Standard IP access list 10
    permit host 192.168.3.3

R3#
```

Step 2: Apply ACL 10 to ingress traffic on the VTY lines.

Use the access-class command to apply the access list to incoming traffic on the VTY lines, on all three routers.

```
R1(config)# line vty 0 4
R1(config-line)# access-class 10 in

R2(config)# line vty 0 4
R2(config-line)# access-class 10 in

R3(config)# line vty 0 4
R3(config-line)# access-class 10 in
```

```
R1>enable
Password:
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line vty 0 4
R1(config-line)#access-list 10 in
                            ^
% Invalid input detected at '^' marker.

R1(config-line)#access-class 10 in
R1(config-line)#
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show run | section line vty
line vty 0 4
 access-class 10 in
 login local
 transport input ssh
R1#
```

```
R2>enable
Password:
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#line vty 0 4
R2(config-line)#access-class 10 in
R2(config-line)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show run | section line vty
line vty 0 4
 access-class 10 in
 login local
 transport input ssh
R2#
```

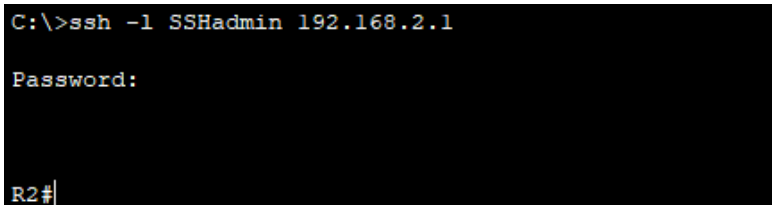
```
R3>enable
Password:
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#line vty 0 4
R3(config-line)#access-class 10 in
R3(config-line)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show run | section line vty
line vty 0 4
 access-class 10 in
 login local
 transport input ssh
R3#
```

Step 3: Verify exclusive access from management station PC-C.

a. Establish an SSH session to 192.168.2.1 from PC-C (should be successful).

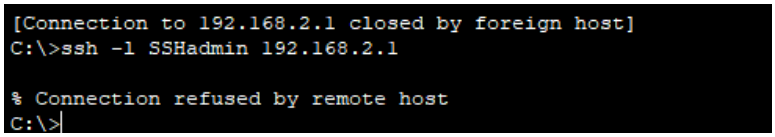
```
PC-C> ssh -l SSHadmin 192.168.2.1
```



Observation / Result: With ACL 10 applied to the VTY lines on R1, R2, and R3, PC-C (192.168.3.3) was still able to establish an SSH session to R2's Lo0 interface, confirming that the management station retains remote access as required.

b. Establish an SSH session to 192.168.2.1 from PC-A (should fail).

```
PC-A> ssh -l SSHadmin 192.168.2.1
```



Observation / Result: After ACL 10 was applied, PC-A (192.168.1.3) attempted to SSH into R2's Lo0 interface and received "% Connection refused by remote host," confirming that only PC-C (192.168.3.3) is permitted remote management access to the routers.

Part 3: Create a Numbered IP ACL 120 on R1

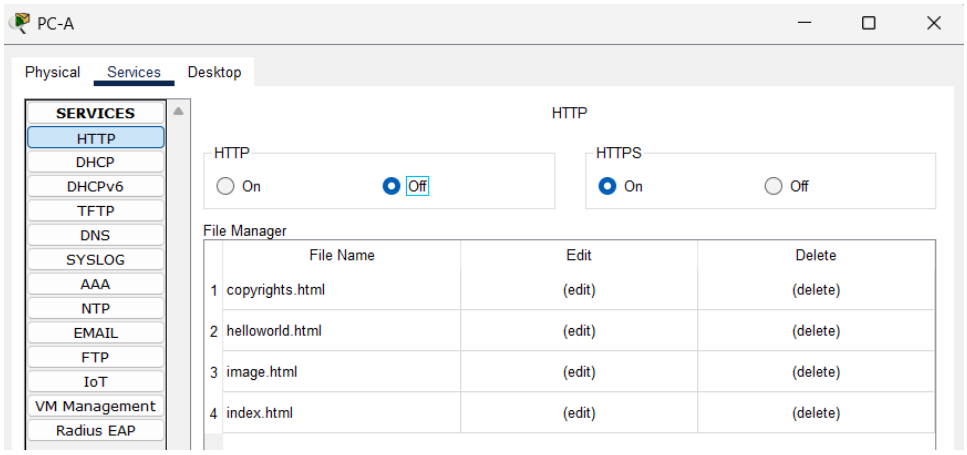
Create an IP ACL numbered 120 with the following rules:

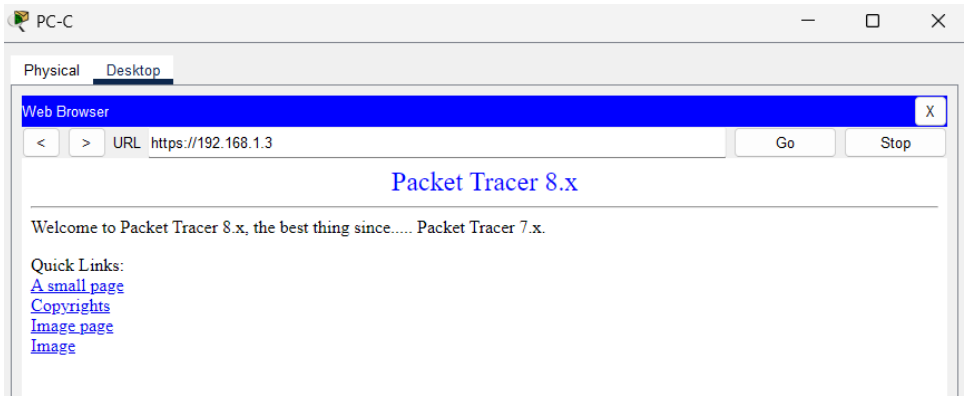
- Permit any outside host to access DNS, SMTP, and FTP services on server PC-A.
- Deny any outside host access to HTTPS services on PC-A.
- Permit PC-C to access R1 via SSH.

Note: Check Results will not show a correct configuration for ACL 120 until it is modified in Part 4.

Step 1: Verify that PC-C can access PC-A via HTTPS using the web browser.

Be sure to disable HTTP and enable HTTPS on server PC-A before testing.





Observation / Result: With HTTP disabled and HTTPS enabled on PC-A, PC-C successfully loaded the server's page over https://192.168.1.3 before ACL 120 was applied, confirming baseline HTTPS connectivity. (An earlier attempt using plain http:// timed out as expected, since the HTTP service had been disabled.)

Step 2: Configure ACL 120 to specifically permit and deny the specified traffic.

```
R1(config)# access-list 120 permit udp any host 192.168.1.3 eq domain
R1(config)# access-list 120 permit tcp any host 192.168.1.3 eq smtp
R1(config)# access-list 120 permit tcp any host 192.168.1.3 eq ftp
R1(config)# access-list 120 deny tcp any host 192.168.1.3 eq 443
R1(config)# access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq
22
```

```
R1>enable
Password:
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 120 permit udp any host 192.168.1.3 eq domain
R1(config)#access-list 120 permit tcp any host 192.168.1.3 eq smtp
R1(config)#access-list 120 permit tcp any host 192.168.1.3 eq ftp
R1(config)#access-list 120 deny tcp any host 192.168.3.3 host 10.1.1.1 eq 22
                                                                    ^
% Invalid input detected at '^' marker.

R1(config)#access-list 120 deny tcp any host 192.168.1.3 eq 443
R1(config)#access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show access-list 120
Extended IP access list 120
    permit udp any host 192.168.1.3 eq domain
    permit tcp any host 192.168.1.3 eq smtp
    permit tcp any host 192.168.1.3 eq ftp
    deny tcp any host 192.168.1.3 eq 443
    permit tcp host 192.168.3.3 host 10.1.1.1 eq 22

R1#
```

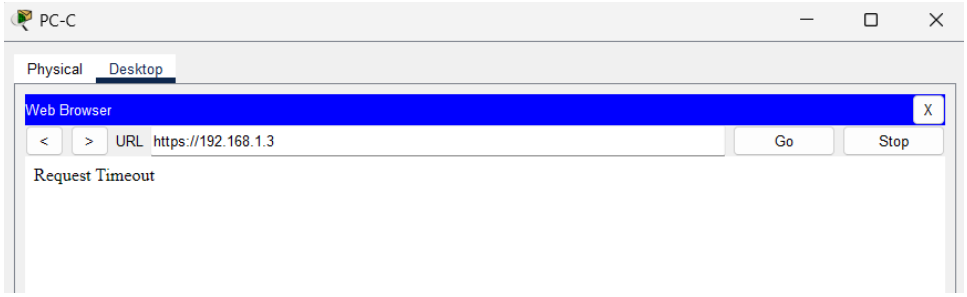
Step 3: Apply the ACL to interface S0/0/0.

```
R1(config)# interface s0/0/0
R1(config-if)# ip access-group 120 in
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#ip access-group 120 in
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip interface s0/0/0 | include access list
  Outgoing access list is not set
  Inbound  access list is 120
R1#
```

Step 4: Verify that PC-C cannot access PC-A via HTTPS using the web browser.



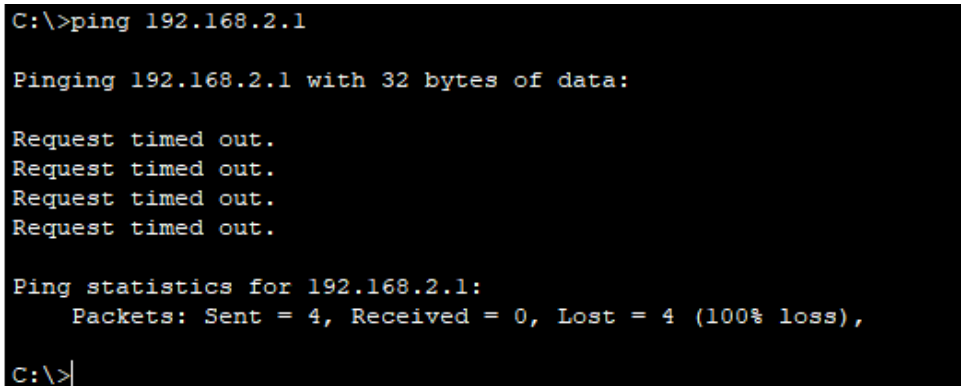
Observation / Result: After ACL 120 was applied to R1's S0/0/0 interface, PC-C's attempt to browse to https://192.168.1.3 resulted in "Request Timeout," confirming that the 'deny tcp any host 192.168.1.3 eq 443' rule is correctly blocking HTTPS traffic to the PC-A server.

Part 4: Modify an Existing ACL on R1

Permit ICMP echo replies and destination unreachable messages from the outside network (relative to R1). Deny all other incoming ICMP packets.

Step 1: Verify that PC-A cannot successfully ping the loopback interface on R2.

```
PC-A> ping 192.168.2.1
```



Observation / Result: Before ACL 120 was modified to permit ICMP, PC-A's ping to R2's Lo0 interface (192.168.2.1) resulted in 4/4 requests timed out (100% loss), since ICMP traffic was implicitly denied by ACL 120's lack of an ICMP permit statement.

Step 2: Make any necessary changes to ACL 120 to permit and deny the specified traffic.

```
R1(config)# access-list 120 permit icmp any any echo-reply
R1(config)# access-list 120 permit icmp any any unreachable
R1(config)# access-list 120 deny icmp any any
R1(config)# access-list 120 permit ip any any
```

```
R1>enable
Password:
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 120 permit icmp any any echo-reply
R1(config)#access-list 120 permit icmp any any unreachable
R1(config)#access-list 120 deny icmp any any
R1(config)#access-list 120 permit ip any any
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show access-lists 120
Extended IP access list 120
    permit udp any host 192.168.1.3 eq domain
    permit tcp any host 192.168.1.3 eq smtp
    permit tcp any host 192.168.1.3 eq ftp
    deny tcp any host 192.168.1.3 eq 443 (12 match(es))
    permit tcp host 192.168.3.3 host 10.1.1.1 eq 22
    permit icmp any any echo-reply
    permit icmp any any unreachable
    deny icmp any any
    permit ip any any

R1#
```

Step 3: Verify that PC-A can successfully ping the loopback interface on R2.

```
PC-A> ping 192.168.2.1
```

```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254
Reply from 192.168.2.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Observation / Result: After ACL 120 was modified to permit ICMP echo-reply and unreachable messages, PC-A's ping to R2's Lo0 interface (192.168.2.1) succeeded with 4/4 replies and 0% loss (avg 1ms), confirming the updated ACL correctly allows return ICMP traffic while still denying other ICMP types.

Part 5: Create a Numbered IP ACL 110 on R3

Deny all outbound packets with a source address outside the range of internal IP addresses on R3.

Step 1: Configure ACL 110 to permit only traffic from the inside network.

```
R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 any
```

```
R3>enable
Password:
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 any
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show access-lists 110
Extended IP access list 110
    permit ip 192.168.3.0 0.0.0.255 any

R3#
```

Step 2: Apply the ACL to interface G0/1.

```
R3(config)# interface g0/1
R3(config-if)# ip access-group 110 in
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface g0/1
R3(config-if)#ip access-group 110 in
R3(config-if)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show ip interface g0/1 | include access list
  Outgoing access list is not set
  Inbound  access list is 110
R3#
```

Part 6: Create a Numbered IP ACL 100 on R3

On R3, block all packets containing a source IP address from the following pool of addresses: any RFC 1918 private addresses, 127.0.0.0/8, and any IP multicast address. Since PC-C is used for remote administration, permit SSH traffic from the 10.0.0.0/8 network to return to host PC-C.

Step 1: Configure ACL 100 to block all specified traffic from the outside network.

Traffic sourced from the internal address space must also be blocked if it is not an RFC 1918 address. In this activity, the internal address space is part of the private address space specified in RFC 1918.

```
R3(config)# access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host
192.168.3.3
R3(config)# access-list 100 deny ip 10.0.0.0 0.255.255.255 any
R3(config)# access-list 100 deny ip 172.16.0.0 0.15.255.255 any
R3(config)# access-list 100 deny ip 192.168.0.0 0.0.255.255 any
R3(config)# access-list 100 deny ip 127.0.0.0 0.255.255.255 any
R3(config)# access-list 100 deny ip 224.0.0.0 15.255.255.255 any
R3(config)# access-list 100 permit ip any any
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#no access-list 100
R3(config)#access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3
R3(config)#access-list 100 deny ip 10.0.0.0 0.255.255.255 any
R3(config)#access-list 100 deny ip 172.16.0.0 0.15.255.255 any
R3(config)#access-list 100 deny ip 192.168.0.0 0.0.255.255 any
R3(config)#access-list 100 deny ip 127.0.0.0 0.255.255.255 any
R3(config)#access-list 100 deny ip 224.0.0.0 15.255.255.255 any
R3(config)#access-list 100 permit ip any any
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show access-lists 100
Extended IP access list 100
    permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3
    deny ip 10.0.0.0 0.255.255.255 any
    deny ip 172.16.0.0 0.15.255.255 any
    deny ip 192.168.0.0 0.0.255.255 any
    deny ip 127.0.0.0 0.255.255.255 any
    deny ip 224.0.0.0 15.255.255.255 any
    permit ip any any

R3#
```

Step 2: Apply the ACL to interface Serial 0/0/1.

```
R3(config)# interface s0/0/1
R3(config-if)# ip access-group 100 in
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface s0/0/1
R3(config-if)#ip access-group 100 in
R3(config-if)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#show ip interface s0/0/1 | include access list
  Outgoing access list is not set
  Inbound  access list is 100
R3#
```

Step 3: Confirm that the specified traffic entering interface Serial 0/0/1 is handled correctly.

a. From the PC-C command prompt, ping the PC-A server. The ICMP echo replies are blocked by the ACL since they are sourced from the 192.168.0.0/16 address space.

```
PC-C> ping 192.168.1.3
```

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Observation / Result: After ACL 100 was applied to R3's S0/0/1 interface, PC-C's ping to PC-A (192.168.1.3) resulted in 4/4 requests timed out (100% loss). This is expected: the ICMP echo-reply from PC-A is sourced from the 192.168.0.0/16 address space, which is explicitly denied by ACL 100's 'deny ip 192.168.0.0 0.0.255.255 any' entry as it re-enters R3 from the outside network.

b. Establish an SSH session to 192.168.2.1 from PC-C (should be successful).

```
PC-C> ssh -l SSHadmin 192.168.2.1
```

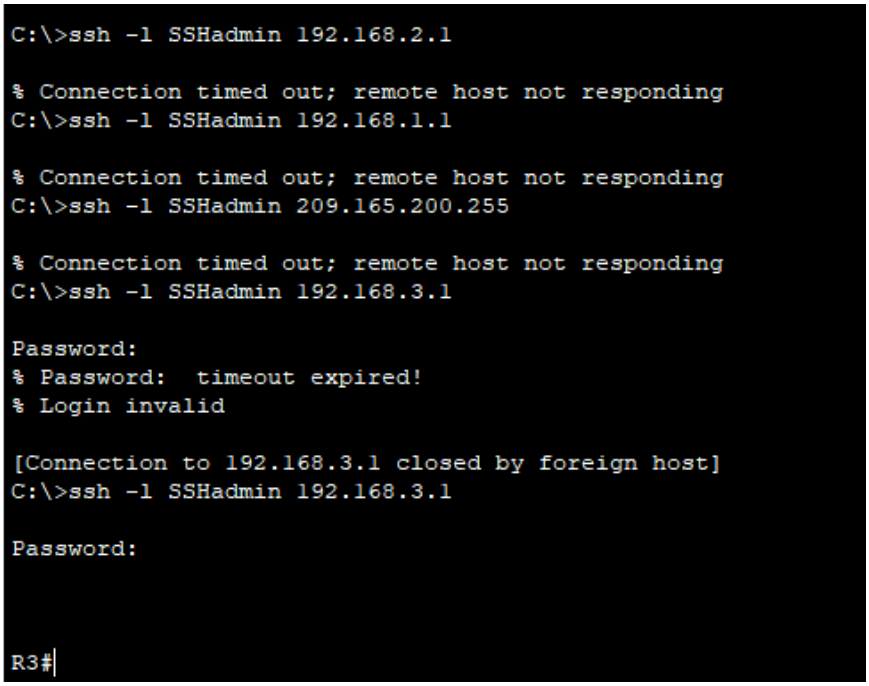
```
C:\>ssh -l SSHadmin 192.168.2.1

% Connection timed out; remote host not responding
C:\>
```

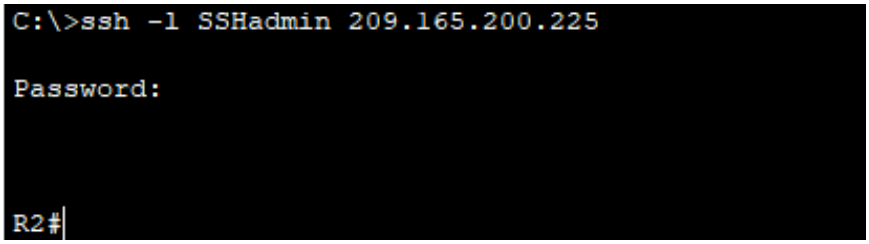
Observation / Result: After ACL 100 was applied to R3's S0/0/1 interface, PC-C's SSH attempt to R2's Lo0 interface (192.168.2.1) resulted in "Connection timed out; remote host not responding." This occurs because ACL 100's SSH exception only permits return traffic sourced from the 10.0.0.0/8 WAN address space (the routers' serial interfaces); R2's Lo0 address falls within 192.168.0.0/16, which is explicitly denied by the ACL, so the SSH reply from 192.168.2.1 is blocked on re-entry to R3. SSH sessions to the routers' actual 10.x WAN interfaces (e.g., R3 S0/0/1 at 10.2.2.1) would succeed under this same ACL.

Supplemental Verification: SSH Access to Router Interfaces (Assessment Checklist)

The Packet Tracer activity's built-in assessment checklist requires establishing SSH sessions from the hosts to R1 G0/1, R2 G0/0 and Lo0, and R3 G0/1, in addition to the specific tests called out in the written procedure above. The following tests were performed from PC-C after all ACLs (10, 120, 100, 110) were fully configured.



Observation / Result: This command history shows four SSH attempts from PC-C: (1) to R2 Lo0 (192.168.2.1) — timed out, consistent with the earlier finding that ACL 100 blocks return traffic from the 192.168.0.0/16 range; (2) to R1 G0/1 (192.168.1.1) — also timed out for the same reason, since 192.168.1.1 likewise falls within the 192.168.0.0/16 range ACL 100 denies on R3's S0/0/1; (3) to 209.165.200.255, a typo for R2 G0/0's actual address (209.165.200.225) — that address is the subnet broadcast address, so it is correctly unreachable; (4) to R3 G0/1 (192.168.3.1) — succeeded, reaching the R3# prompt after an initial password-entry timeout, since this traffic stays local to R3's LAN segment and is not subject to ACL 100.



Observation / Result: After correcting the typo to the actual address 209.165.200.225, the SSH session from PC-C to R2's G0/0 interface succeeded, reaching the R2# prompt. This is expected: 209.165.200.225 is a public address outside all of ACL 100's deny ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 127.0.0.0/8, 224.0.0.0/4), so the return traffic falls through to the final "permit ip any any" statement.

Note: PC-A could not be tested against this checklist item because ACL 10 (applied in Part 2) restricts VTY access on all three routers to PC-C (192.168.3.3) only — this is the intended, correct behavior of the configuration, not an oversight.

Conclusion / Reflection

This activity confirmed that baseline connectivity existed between all key devices before any security controls were introduced, establishing a working network as the starting point. ACL 10 was then applied to the VTY lines on R1, R2, and R3 to restrict remote management access to the designated station, PC-C, which was verified by successful SSH from PC-C and a refused connection from PC-A. On R1, ACL 120 was built to protect the PC-A server by permitting only the specific services it needed to expose to the outside (DNS, SMTP, and FTP) while explicitly denying HTTPS, and was later refined to allow diagnostic ICMP traffic (echo-replies and unreachables) without opening ICMP broadly. On R3, ACL 110 enforced that only traffic genuinely originating from the 192.168.3.0/24 network could leave the internal LAN, and ACL 100 added anti-spoofing protection on the WAN-facing serial interface by blocking inbound traffic claiming to originate from private (RFC 1918), loopback, or multicast address space, with a narrow exception preserved for legitimate SSH management traffic.

A few configuration mistakes came up along the way and served as useful reminders of ACL syntax and behavior: the access-class command (applied on VTY lines) was initially confused with access-list (used to define the ACL itself); a typo in the multicast address for ACL 100 (244.0.0.0 instead of 224.0.0.0) was caught by comparing the show access-lists output against the intended rule; and an attempt to remove a single malformed entry from a numbered ACL using the no access-list 100 <entry> syntax unexpectedly cleared the entire access list, requiring it to be rebuilt from scratch. Each issue was identified by carefully reviewing command output rather than assuming success, reinforcing the importance of verifying configuration state after every change rather than only after applying an ACL to an interface.